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Research Article

# Benchmarking Hydrogen and Oxygen Electrocatalytic Reactions in Acidic and Alkaline Media on Iridium Single-Crystal Surfaces

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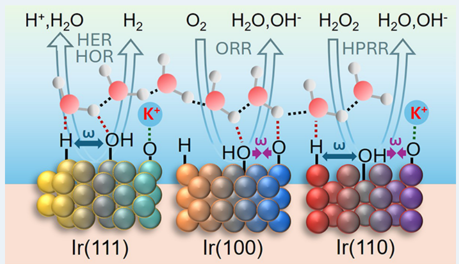
**ABSTRACT:** Material composition, surface structure, adsorbate coverage, electrolyte composition, and pH are all factors that play a critical role in the reactivity of the electrochemical interface. Iridium, despite its importance for energy conversion technologies, remains largely underexplored as an electrocatalyst in its metallic surface state. Here, we benchmark the hydrogen evolution and oxidation reactions (HER/HOR), oxygen reduction reaction (ORR), and hydrogen peroxide reduction reaction (HPRR) on Ir(111), Ir(100), and Ir(110) single crystals in acidic ($HClO_4$) and alkaline (KOH) media using RDE/RRDE methods. Across both pH conditions, all reactions exhibit strong dependency on surface orientation but are not monotonically dependent on surface defects or lower surface site coordination number. For HER/HOR, Ir(111) is the most active facet in both electrolyte media, lowering the water dissociation kinetic barrier more effectively than on Ir(110). In contrast, Ir(110) is the most active for ORR in both pH conditions, whereas selectivity is strongly pH- and potential-dependent. Hydrogen peroxide yields are below 1% in acidic media but rise dramatically in alkaline electrolytes, reaching very high values (>50%) near open-circuit conditions. The impact of co-adsorption between $O_{ad}$ and $OH_{ad}$ species shows a strong correlation to $H_2O_2$ production in alkaline media, suggesting water bond-breaking kinetics is an important parameter directing ORR selectivity. Additionally, HPRR trends indicate that in acidic media, ORR does not necessarily proceed through $H_2O_2$ intermediates, as low HPRR activity is not accompanied by $H_2O_2$ accumulation during ORR at $E > 0.4$ V. In contrast, in alkaline media, $H_2O_2$ accumulation correlates with reduced HPRR activity. The influence of surface sites and pH can be rationalized in terms of adsorbate speciation, lateral interactions, and interfacial water dynamics. Our results offer benchmark reactivity data on Ir(hkl) surfaces and across pH that complements literature results on Pt and Au surfaces while solidifying the role of interfacial water processes in determining the functional properties of electrochemical interfaces in aqueous environments.




**KEYWORDS:** *iridium single crystals, electrocatalysis, pH effect, hydrogen evolution reaction, hydrogen oxidation reaction, oxygen reduction reaction, hydrogen peroxide reduction reaction*

## ■ INTRODUCTION

Fundamental studies on single-crystal electrodes provide a critical foundation for understanding electrocatalytic reactions at the atomic level. In contrast to nanoparticles and polycrystalline materials, whose structural heterogeneity complicates mechanistic interpretation, single-crystal surfaces offer well-defined atomic arrangements that enable the most direct correlation of surface structure with adsorption, oxide formation, and reaction kinetics. This approach has been demonstrated extensively for Pt surfaces, where both trends across basal planes and controlled defects across stepped crystals highlight the complex interactions between surface sites, adsorbate species, and water at the double layer.[1−5] For iridium, however, only a limited number of reports are available on the reactivity trends across its single-crystal facets. Our recent contribution relied on CO-displacement, in situ SHINERS, and DFT to present a full characterization of the dynamics of adsorption on Ir(hkl) in acidic media and the role of these species in electrocatalysis.[6] By identifying how specific surface site features influence elementary reaction steps, single-crystal studies establish rigorous structure-reactivity relationships that are essential for interpreting the behavior of practical catalysts and for guiding the rational design of more active and efficient Iridium-based electrocatalysts.

Water cycle reactions such as the hydrogen evolution reaction (HER), hydrogen oxidation reaction (HOR), and oxygen reduction reaction (ORR), are central to a wide range of energy conversion and storage technologies.[7−12] While the oxygen evolution reaction (OER) is also an important process, it takes place on oxide surfaces, quite distinct from a metallic surface

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of interest in this work. Hydrogen evolution reaction and hydrogen oxidation reaction are especially valuable as prototypical electrocatalytic processes for probing interfacial kinetics on well-defined surfaces. Despite their apparent simplicity, both HER and HOR are highly sensitive to surface structure, adsorbate energetics, and electrolyte environment, making them ideal model reactions for establishing fundamental structure-activity relationships on well-defined iridium surfaces. The ORR, on the other hand, is a particularly complex reaction, involving four electrons, with variable selectivity toward water via a 4e$^-$ pathway or hydrogen peroxide ($H_2O_2$) via a 2e$^-$ pathway. The generation of $H_2O_2$ during ORR has a critical role in the chemical degradation of system components such as proton exchange membranes (PEMs) and catalyst supports.[13−17] Although in the last few decades, most investigations regarding fuel cell electrodes have focused on the selectivity of ORR at the cathode, peroxide formation is also relevant at the anode during startup and shutdown, when oxygen crossover is most relevant. In this condition, Pt-group catalysts can promote a two-electron ORR pathway, leading to localized production of $H_2O_2$,[18,19] where it can either decompose into reactive oxygen species (ROS),[20] including hydroxyl radicals ($^{\cdot}$OH), via Fenton-type reactions, or diffuse into the membrane and trigger structural degradation.[21,22] Therefore, control over both peroxide formation and reduction is essential to mitigate chemical degradation and maintain the long-term stability of electrochemical devices.[23]

This work establishes clear facet- and pH-dependent trends in the electrocatalytic behavior of Ir(111), Ir(100), and Ir(110) single-crystal electrodes toward HER, HOR, ORR, and HPRR. We found that Ir(111) is the most active surface for hydrogen electrocatalysis in both acidic and alkaline media, whereas Ir(110) is the most active for ORR, highlighting an intrinsic trade-off between optimal hydrogen and oxygen reduction activity. Both HER and HOR are strongly suppressed in alkaline media, consistent with the added kinetic barrier associated with water dissociation and interfacial water reorganization, while ORR activity shows comparatively weak pH dependence but pronounced changes in peroxide selectivity. In acidic media, ORR proceeds with negligible $H_2O_2$ formation on all Ir(*hkl*) facets, whereas in alkaline media, peroxide generation increases substantially and depends strongly on surface orientation. A comparison of ORR and HPRR further suggests that in acidic media, ORR does not necessarily proceed through $H_2O_2$ as an obligatory intermediate, while in alkaline media, peroxide accumulation is consistent with inhibited HPRR at higher potentials. Overall, these results demonstrate that iridium electrocatalysis is governed not only by adsorption energetics, but also by facet-dependent interfacial structure, surface charge, and adsorbate coverage.

## ■ RESULTS AND DISCUSSION

### Surface Characterization in Acidic and Alkaline Media

We begin by establishing the quality of the surfaces using atomic force microscopy (AFM). Figure 1a–c reveals the surface morphology of the pristine Ir(111), Ir(100), and Ir(110) surfaces. The Ir(111) (Figure 1a) surface exhibits well-resolved 150–200 nm wide atomically flat terraces separated by monatomic steps of approximately 0.22 nm in height, consistent with the theoretical interlayer spacing of 0.222 nm for the fcc Ir(111) surface. The Ir(100) surface (Figure 1b) reveals exceptionally broad terraces with a pronounced step edge of approximately 1 nm. The

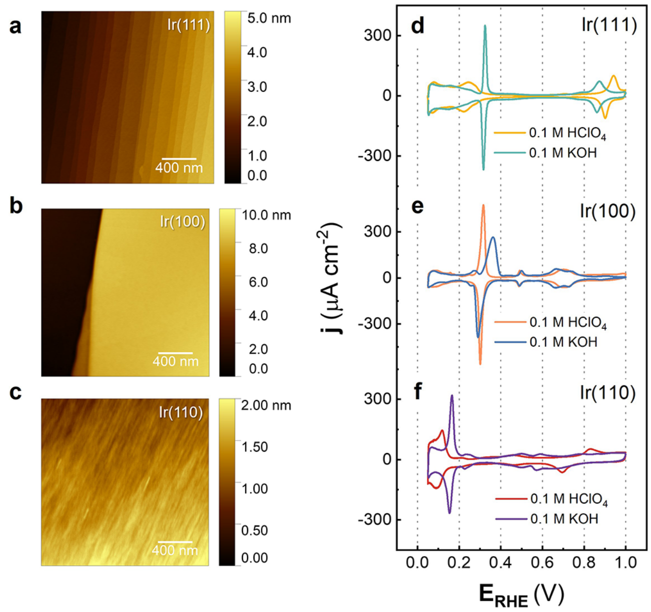


**Figure 1.** AFM images and cyclic voltammograms of pristine (a, d) Ir(111), (b, e) Ir(100), and (c, f) Ir(110) in 0.1 M $HClO_4$ (yellow, orange, and red solid lines), and in 0.1 M KOH (green, blue, and purple solid lines), at 0.05 V s$^{-1}$

Ir(110) surface (Figure 1c) displays a comparatively rougher topography with step features on the order of 0.1 nm and a characteristic undulating texture. Although these features are consistent with the reported tendency of fcc(110) surfaces toward missing-row reconstruction,[24] the present measurements do not allow us to conclusively confirm reconstruction on Ir(110). Additionally, the pristine quality of all three electrodes is remarkably high, as evidenced by the laterally extensive and atomically resolved terrace-step structures spanning the full 2 μm field of view, the absence of adsorbate-related protrusions or pit defects, and the minimal root-mean-square roughness across each scan area.

Figure 1d–f shows the voltammetric behavior of fresh annealed single-crystalline iridium surfaces Ir(111), Ir(100), and Ir(110) in both acidic and alkaline media, using 0.1 M $HClO_4$ and 0.1 M KOH as supporting electrolytes, respectively. Note that all electrode potentials reported here are referenced versus a reversible hydrogen electrode (RHE), and to avoid surface history effects (via iridium oxide formation), all the data in this investigation were collected in the positive-going scan, starting at 0 V, and in descending order of rotation rate (for reactions). In acidic solution (0.1 M $HClO_4$), where perchlorate ions do not specifically adsorb, the observed surface processes are primarily governed by water-derived species, such as adsorbed hydrogen ($H_{ad}$), adsorbed hydroxyl ($OH_{ad}$), and adsorbed oxygen ($O_{ad}$), in line with recent literature.[6] In alkaline media (0.1 M KOH), although the adsorbed species are the same as observed in acidic environments, hydroxyl ions are abundantly available, leading to observable changes in the voltammetric profiles that reflect modifications in adsorption and surface oxidation behavior.

For Ir(111) (Figure 1d), the voltammogram in acidic media exhibits a broad feature between 0.05 and 0.15 V, attributed to hydrogen desorption/adsorption, followed by a distinct peak near 0.20 V, which is assigned to the co-adsorption of hydrogen and hydroxyl species.[6] At more positive potentials, a broad oxidation feature emerges, with a peak near 0.8 V associated with




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the formation of adsorbed oxygen species. These features are consistent with prior studies and reflect the well-characterized adsorption process on Ir(111) in acidic media.[25–31] In alkaline media, the $H_{ad}$ region remains largely unaltered. However, the second process between 0.19 and 0.30 V, related to $OH_{ad}$ adsorption as identified in acidic media, is shifted to more positive potentials and is characterized by a sharp adsorption and desorption peak. This assignment is supported by charge displacement experiments (see Figure S1) and aligns with previous reports in the literature.[32,33] Interestingly, subsequent surface oxidation occurs at lower potentials in alkaline than acid media, ca. 0.70 V, indicating that surface oxidation processes are facilitated under alkaline conditions. This behavior can be explained by a combination of the increased availability of hydroxide ions and the existence of noncovalent interactions with alkali cations at the double layer.[34] The higher hydration energy for proton species (hydronium) compared to alkali cations such as $Li^+$, $Na^+$, and $K^+$ will have a direct impact on the stability of the $OH_{ad}$ layer, hence a positive $OH_{ad}$ formation shift in alkaline versus acidic media. As such, the conversion of $OH_{ad}$ to $O_{ad}$ requires additional energy in acidic than alkaline media, similarly to what has been observed on Pt(111) electrodes.[34,35] These observations highlight the significant influence of the electrolyte on the adsorption and oxidation of iridium surfaces.

For Ir(100), shown in Figure 1e, voltammograms in both acidic and alkaline media show a clear $H_{ad}$ feature below 0.23 V (see more details in Figure S1). Furthermore, in acidic media, the sharp peak at 0.29 V is primarily attributed to $OH_{ad}$, with a small contribution from co-adsorbed $H_{ad}$. In alkaline media, this peak becomes broader and asymmetric, which demonstrates that despite the abundance of $OH^-$ species, the adsorption kinetics are slower. Moreover, a small peak at 0.46 V (in both alkaline and acidic media) corresponds to the onset formation of $O_{ad}$ [26,28,30,36] which shows a similar onset potential for both electrolytes, but it is less reversible in alkaline media. Notably, the last adsorption process, centered at 0.70 V and associated with a secondary oxidation process, exhibits a lower onset potential in KOH compared to $HClO_4$. However, in contrast with Ir(111), the kinetics of the $O_{ad,}$ particularly evident from peak separation, remain similar for both acidic and alkaline media for Ir(100).

Lastly, for Ir(110), regardless of the electrolyte pH, there is no clear $H_{ad}$ region (Figure 1f). Instead, a potential region below 0.17 V corresponds to the co-adsorption of hydrogen and hydroxyl species, which is a result of its lower $E_{pztc}$ compared with the other low-index facets (see details in Figures S1 and S2). Additionally, in acidic media, a broad feature centered at 0.11 V is attributed to $OH_{ad}$ species, while smaller peaks observed at potentials above 0.4 V, including an irreversible peak at 0.84 V (positive scan), are associated with oxide formation.[26,28,30] In contrast, in the presence of KOH, the $OH_{ad}$ adsorption peak is sharper and shifted to higher potentials (0.16 V) compared to the acidic environment. Furthermore, the oxidation features, including those associated with $O_{ad}$, are more reversible in KOH than in $HClO_4$ and seem to occur also at lower potentials.

Additionally, Figure S3 shows the total charge curves, revealing that the potential of zero total charge ($E_{pztc}$, electrode potential where the net surface charge density on the interface is zero) is positively shifted in alkaline relative to acidic media for all iridium surfaces, $E_{pztc,111}$ = 0.20 V, 0.29 V, $E_{pztc,100}$ = 0.29 V, 0.34 V, and $E_{pztc,110}$ = 0.10 V, 0.12 V (versus RHE), in 0.1 M $HClO_4$ and KOH, respectively. $E_{pztc,111}$ value agrees with previous work[37] that demonstrates that the potential of maximum entropy in 0.1 M $HClO_4$ is below 0.3 V (RHE). (For a detailed calculation of $E_{pztc}$, refer to Supplementary Note 1 and Table S1). This shift indicates a change in interfacial composition and implies that, when comparing different surfaces or electrolytes within the same potential region, adsorbate coverage and adsorption behavior will differ. As a result, the interfacial water structure and stabilization of reactive intermediates can change, which is expected to affect catalytic behavior in HER, HOR, ORR, and HPRR. With this picture of the nature and distribution of surface species on Ir(*hkl*) now established, we examine key electrochemical reactions taking place within the Ir "metallic" surface state: the hydrogen evolution reaction, hydrogen oxidation reaction, oxygen reduction reaction, and hydrogen peroxide reduction reaction. Note that the results in Figure 1 make it clear that up to 1 V, although surface oxidation is evident with the formation of $O_{ad}$ in all three low-index facets, the coverage values are still constrained to a surface monolayer, rendering it quite distinct from $IrO_x$ surfaces that are present during the oxygen evolution reaction. As such, by limiting our reactions of interest to the electrode potential window where surface-adsorbed species are constrained, we can explore their role in dictating Ir surface reactivity.

## Hydrogen Evolution Reaction and Hydrogen Oxidation Reaction (HER and HOR)

On a freshly annealed electrode, all three Ir single-crystal surfaces exhibit rapid HER kinetics in acidic media (Figure 2a and eq 1). At 10 mA $cm^{-2}$, the overpotential trend follows Ir(111) < Ir(110) < Ir(100), with values of 23, 30, and 34 mV, respectively. As discussed in our previous study,[6] these small facet-dependent differences are consistent with modest variations in hydrogen adsorption and surface coverage. In alkaline media, however, HER becomes markedly slower, as expected from the additional requirement for water dissociation,[38] while the facet dependence becomes much more pronounced (Figure 2b and eq 4). Ir(111) remains the most active surface, reaching 10 mA $cm^{-2}$ at 102 mV, whereas Ir(110) and Ir(100) require 293 and 334 mV, respectively. The comparison across pH therefore shows that all Ir facets are penalized in an alkaline electrolyte, with the largest loss observed for Ir(100) (Figure 2c).

**Hydrogen evolution reaction, in acidic media**:

$$H^+ + e^- \rightarrow H_{opd} \text{ (Volmer Step)} \quad (1)$$

$$H_{opd} + H^+ + e^- \rightarrow H_2 \text{ (Heyrovsky Step)} \quad (2)$$

$$H_{opd} + H_{opd} \rightarrow H_2 \text{ (Tafel Step)} \quad (3)$$

**Hydrogen evolution reaction, in alkaline media:**

$$H_2O + e^- \rightarrow H_{opd} + OH^- \text{ (Volmer Step)} \quad (4)$$

$$H_{opd} + H_2O + e^- \rightarrow H_2 + OH^- \text{ (Heyrovsky Step)} \quad (5)$$

$$H_{opd} + H_{opd} \rightarrow H_2 \text{ (Tafel Step)} \quad (6)$$




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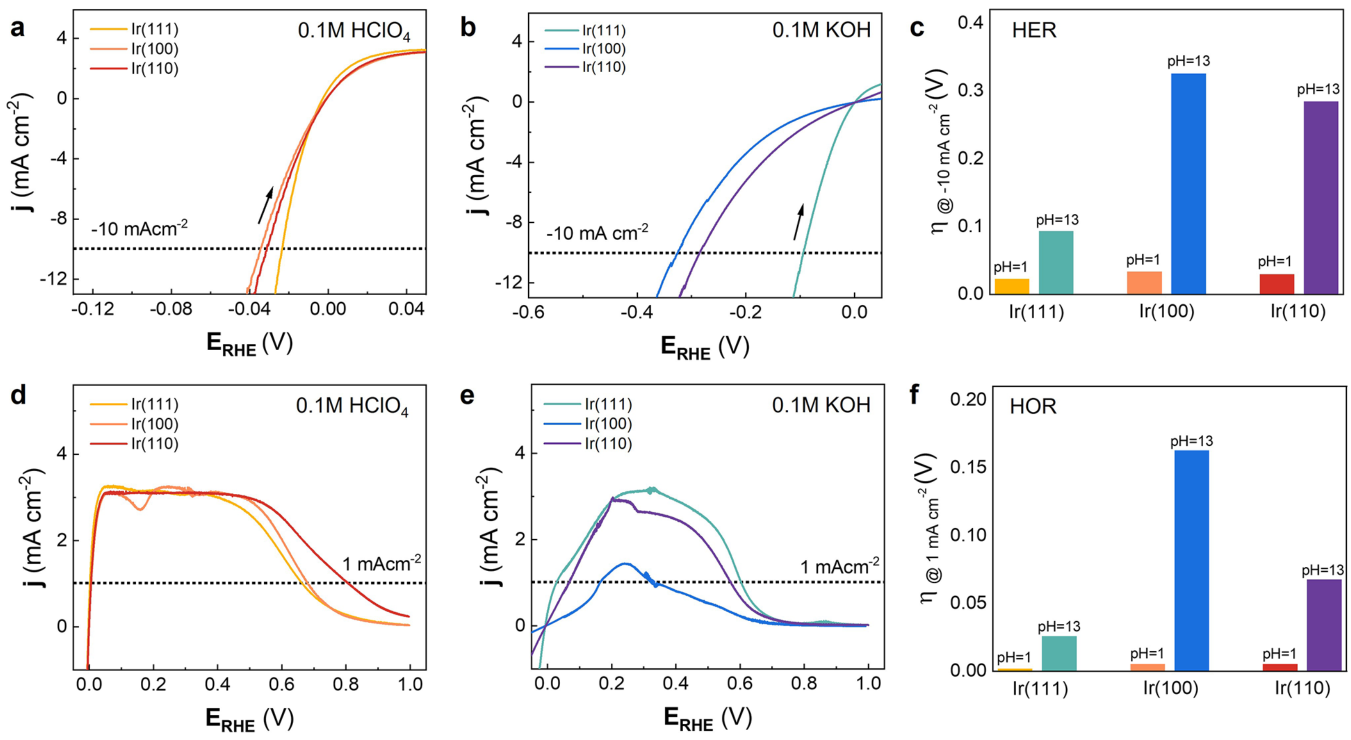


**Figure 2.** (a, b) HER polarization curves in 0.1 M $HClO_4$ and 0.1 M KOH, respectively, (c) bar graph comparing HER overpotentials in acidic and alkaline media estimated from the kinetic current = 1 mA $cm^{-2}$ from Figure S6. (d, e) HOR polarization curves in 0.1 M $HClO_4$ and 0.1 M KOH, (f) bar graph comparing HOR overpotentials in acidic and alkaline media. All measurements were carried out on Ir(111), Ir(100), and Ir(110) surfaces at a scan rate of 0.01 V $s^{-1}$ and a rotation speed of 2500 rpm.

For HER on Ir(*hkl*), the measured polarization response is strongly influenced by non-kinetic contributions. Under limited mass transport, concentration overpotentials increase significantly (Figure S4), and even at high rotation rates, interfacial $H_2$ supersaturation and bubble formation introduce additional uncompensated resistance that obscures a well-defined Tafel region (Figure S5). Consequently, the Tafel slope cannot be extracted reliably from the raw polarization curves alone. To mitigate this limitation, we analyzed differential Tafel plots (Figure S5), which provide a more robust estimate of the apparent slope over a broader potential window, as demonstrated previously.[39,40] This analysis indicates a pH-dependent shift in the rate-limiting step toward the initial charge-transfer event: in acidic media, Ir(*hkl*) exhibits an apparent slope of ∼30 mV $dec^{-1}$, consistent with Tafel recombination as the dominant kinetic bottleneck, whereas in alkaline media the slope increases to ∼120 mV $dec^{-1}$, consistent with a Volmer-limited pathway, in agreement with previous observations on Pt(111) in alkaline media.[41]

The HOR trends mirror those observed for HER. In acidic media, all three surfaces display extremely low overpotentials considering the kinetic current at 1 mA $cm^{-2}$ (Figure S6), namely, 2 mV for Ir(111) and 5.5 mV for both Ir(110) and Ir(100), confirming very fast $H_2$ oxidation kinetics (Figure 2d). Because HOR is intrinsically rapid in this regime, the accessible overpotential window before the onset of mass-transport limitation is very narrow. In alkaline media, however, HOR is strongly inhibited and becomes highly structure-sensitive (Figure 2e). Ir(111) remains the most active surface, requiring only 26 mV to reach 1 mA $cm^{-2}$, followed by Ir(110) at 68 mV and Ir(100) at 163 mV (Figure 2f). Notably, Ir(100) does not exhibit a well-defined diffusion-limited current plateau, suggesting partial blockage of active sites, whereas Ir(110) shows two distinct plateau regions.

Notably, the decline in HOR activity at $E > 0.4$ V does not correlate directly with the conversion of $OH_{ad}$ to $O_{ad}$. In acidic media, the potential at which HOR begins to decay follows the order Ir(111) < Ir(100) < Ir(110), which does not mirror the sequence for $O_{ad}$ formation. In alkaline media, HOR decay likewise begins well before the onset of iridium oxide formation. Together, these observations indicate that the loss of activity cannot be explained solely by a simple surface oxide species acting as spectators. Rather, it likely reflects the combined effects of interfacial water dynamics, which can hinder $H_2$ access to active sites, and lateral interactions among adsorbates, which modify site availability and adsorption energetics. For HER, the dominant lateral effect is expected to be H–H interaction because the reaction proceeds in the H-covered region. For HOR, both H–H and H–OH interactions become relevant, since $H_2$ oxidation extends into a potential window where $OH_{ad}$ is present. Under these conditions, $OH_{ad}$ can either block sites required for $H_2$ dissociation or modify hydrogen adsorption energetics through lateral H–OH interactions. To account for these coverage-dependent effects, we next analyze the interaction between adsorbates using a microkinetic model.

Microkinetic analysis using a Frumkin isotherm (Figure S7 and Supplementary Note 2) reveals repulsive H–H interactions on all Ir(*hkl*) facets in both acidic and alkaline media (Table S2). In acidic, the repulsion decreases in the order Ir(111) > Ir(100) > Ir(110), with $\omega_{111} = -13.6$ kJ $mol^{-1}$,




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$\omega_{100}$ = −7.4 kJ mol$^{-1}$, and $\omega_{110}$ = −2.5 kJ mol$^{-1}$. In alkaline media, the order changes to Ir(100) > Ir(111) > Ir(110), with $\omega_{100}$ = −15.0 kJ mol$^{-1}$, $\omega_{111}$ = −5.0 kJ mol$^{-1}$, and $\omega_{110}$ = 0 kJ mol$^{-1}$, indicating negligible H–H interaction on Ir(110) near the HER onset. These repulsive interactions are expected to limit the maximum hydrogen coverage and weaken H adsorption as coverage increases. However, the extracted H-H repulsion on Ir is weaker than that reported for Pt(111), suggesting that coverage-induced penalties are less severe on Ir and therefore less detrimental to HER kinetics.

For HOR, the H–OH interaction is likewise repulsive but much more facet- and pH-dependent. In acidic media, the interaction follows $\omega_{111}$= −149, $\omega_{110}$ = −12.4, and $\omega_{100}$ = 0 kJ mol$^{-1}$. Although these values suggest substantial differences in co-adsorption energetics, their kinetic consequences cannot be resolved clearly in acidic media because the current becomes transport-limited before intrinsic surface effects can be fully separated (Figure 2d). In alkaline media, by contrast, facet dependence is directly reflected in the activity. The lower HOR activity of Ir(100) relative to Ir(111) and Ir(110) is consistent with its much stronger repulsive H–OH interaction, ω = −250 kJ mol$^{-1}$, which disfavors the coexistence of H and OH on the surface. In contrast, ω = 0 kJ mol$^{-1}$ for Ir(111) and Ir(110), consistent with their superior HOR activity (Figure 2e). This interpretation is supported by CO-displacement measurements (Figures S1 and S2), which show that at $E$ > 0.25 V, Ir(111) is predominantly OH-covered, Ir(100) exhibits mixed H/OH coverage, and Ir(110) is covered mainly by OH/O species. Together, these results indicate that HOR activity in alkaline media is governed primarily by the balance between $H_{ad}$ and $OH_{ad}$ co-adsorption, rather than by hydrogen binding alone.

The influence of adsorbate lateral interactions is not the only factor contributing to the facet-dependent HER and HOR activity trends. Four other factors should also be considered as follows. First, low-coverage DFT calculations by Haijun and co-workers[42] show that the intrinsic hydrogen adsorption strength follows Ir(111) < Ir(110) < Ir(100), with adsorption free energies of −45.4, −55.0, and −72.4 kJ mol$^{-1}$, respectively, at θ = 0.06 (Details in Table S3 and Supplementary Note 3). This trend is broadly consistent with the HER data, as stronger H binding on Ir(100) and Ir(110) would disfavor hydrogen recombination and desorption relative to Ir(111). Second, the effect of lateral repulsion becomes increasingly important at higher coverage and can weaken hydrogen binding in a facet-dependent manner, particularly on Ir(111), where DFT also predicts a coverage-driven shift in the preferred adsorption site. Third, the hydrogen coverage measured under reaction conditions is not necessarily an equilibrium quantity. A faster Volmer replenishment on Ir(111), for example, could sustain a higher reactive H population despite stronger lateral repulsion. Finally, interfacial solvation and double-layer structure are also likely to contribute. Raman measurements[6] on Ir($hkl$) show potential-dependent restructuring of interfacial water in the same potential region where HOR activity declines, indicating that changes in interfacial water organization and field strength may further constrain coupled proton/electron transfer, particularly in alkaline media. However, applying SHINERS under highly alkaline media remains technically challenging, as demonstrated in a previous investigation.[43]

A useful benchmark is Pt($hkl$), for which alkaline HER/HOR also exhibits pronounced facet dependence, but with a different order: Pt(110) > Pt(100) > Pt(111). In the present study, the most active Ir facets are approximately fivefold more active than the best Pt facets under comparable conditions. More importantly, the different facet-activity trends on Pt and Ir show that surface geometry alone is insufficient to predict reactivity across fcc metals. The higher HER activity of Ir relative to Pt likely does not arise from a different elementary mechanism, but rather from a more favorable balance among H binding, coverage, and lateral interactions. In particular, the weaker H-H repulsion on Ir reduces the coverage-induced penalty on hydrogen adsorption in the HER region. By contrast, HOR is more strongly shaped by facet-dependent H–OH interactions, with OH adsorption introducing a pronounced site-blocking penalty, especially in alkaline media.

Investigations using single-crystalline surfaces provide a rigorous basis for comparing intrinsic electrocatalytic activity due to high atomic-level control. In contrast, nanoparticle catalysts expose a distribution of facets, defects, and coordination environments, such that the measured activity represents an average over sites with different reactivities. For instance, prior studies[44] using metallic nanoparticles concluded lower activity for iridium in comparison with platinum. Our results show that electrocatalysis is strongly structure-sensitive, with Ir(111) substantially more active than Ir(100) and Ir(110). Therefore, the lower apparent activity of iridium nanoparticles likely reflects the contribution of less active facets, demonstrating that nanoparticle investigations alone cannot resolve site-specific electrocatalytic activity.

### Oxygen Reduction Reaction and Hydrogen Peroxide Reduction Reaction (ORR and HPRR)

All ORR polarization curves present a diffusion-limited current density, $j_d$, $E$ < 0.25 V, followed by a region with mixed control (diffusion and kinetic), 0.25 < $E$ < 0.80 V; details in Figure 3a,b. ORR on Ir(100) does not exhibit a well-defined diffusion-limited plateau in either acidic or alkaline media, but rather a plateau-like region with a suppressed current density. This behavior suggests that, in addition to mass transport, partial blockage of the surface by co-adsorbed species affects both HOR and ORR current by reducing the number of available active sites. If this blocking contribution is only weakly dependent on rotation rate, then the observed linear dependence of the current density at 0.1 V on $\omega^{1/2}$ (Figure S8) supports the use of the Koutecký–Levich relation[45] as an approximate means to extract $j_k$ for comparative purposes. On this basis, the kinetic current density was estimated at 0.1 V, and the results are presented in Figure S9. In acidic media, the overpotentials at a kinetic current density of −4 mA cm$^{-2}$ follow the order: Ir(110) (640 mV) < Ir(100) (700 mV) < Ir(111) (814 mV), indicating that Ir(110) is a slightly more active surface for ORR (Figure 3c). The trend observed can be attributed to the higher density of low-coordination sites on Ir(110), which would facilitate oxygen adsorption and dissociation, critical steps in the ORR mechanism.[6] Conversely, Ir(111), with its close-packed structure, exhibits the highest overpotential, likely due to weaker oxygen binding on terrace sites. The same trends in overpotential as in acid are observed in alkaline media: Ir(110) (679 mV) < Ir(100) (714 mV) < Ir(111) (806 mV), as shown in Figure 3b, which suggests that the intrinsic surface properties of Ir($hkl$) play a majority role in governing the activity rather than the electrolyte. For reference, ORR polarization curves at different rotation rates are shown in Figure S10 for




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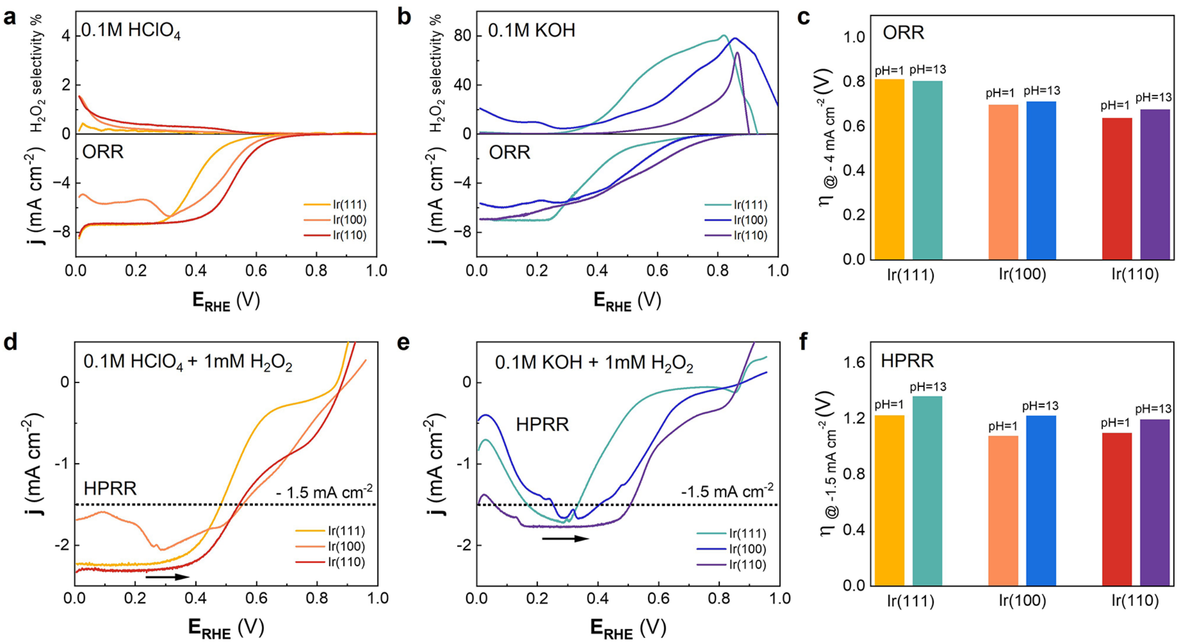


**Figure 3.** (a, b) Oxygen reduction reaction (ORR) polarization curve and hydrogen peroxide selectivity in $O_2$ saturated 0.1 M $HClO_4$ and 0.1 M KOH, respectively, (c) bar graph comparing ORR overpotentials in acidic and alkaline media, estimated from the kinetic current = −4 mA cm$^{-2}$ from Figure S9. (d, e) Polarization curves recorded during the hydrogen peroxide reduction reaction (HPRR), (f) bar graph comparing HPRR overpotentials, in acidic and alkaline media, respectively, with the addition of 1 mM $H_2O_2$. All data were collected at a scan rate of 0.01 V s$^{-1}$ and a rotation speed of 2500 rpm.

both acidic and alkaline media. These results are in line with previous investigations[32,46] showing that iridium, as either single crystals or nanoparticles, exhibits lower ORR activity than Pt-based catalysts, despite the similarity of the Ir and Pt d-band centers. This suggests that the d-band center alone does not fully account for the observed catalytic behavior, but rather interfacial speciation, particularly interfacial water, plays an important role along with the surface electronic properties. Additionally, avoiding surface oxidation is particularly significant for ORR on Ir(100), which exhibits pronounced hysteresis between the positive- and negative-going scans. This effect is most evident on a freshly prepared surface at 2500 rpm and becomes less pronounced upon cycling, particularly at lower rotation rates. Hysteresis is also greater in acidic than in alkaline media, with an unexpectedly higher current density observed in the negative-going scan (Figure S10b,e, dotted lines). This behavior is more likely associated with the accumulation of defect sites on the (100) terraces than with the formation of oxygenated or oxide-like species, which would be expected to suppress the current density.

It is noteworthy that ORR activity exhibits only a weak pH dependence (especially compared with HER/HOR), whereas ORR selectivity shows a much stronger pH sensitivity. Using the rotating ring disk electrode (RRDE) method, it is possible to determine the selectivity of the ORR (top panel of Figure 3a,b) based on the two $O_2$ reduction pathways (see eqs 10–13). Ir(*hkl*) present a high selectivity for the 4-electron pathway during ORR, with $H_2O_2$ selectivity below 1% in acidic media, in agreement with previous work,[47] in sharp contrast to results on Pt(111) where $H_2O_2$ selectivity reaches 55% for $E < 0.2$ V.[48] Our hypothesis is that Ir(*hkl*) generates much less $H_2O_2$ than Pt in acidic media because $H_{ad}$ on Ir(*hkl*) exhibits weaker lateral repulsion [Ir(111), Ir(100), and Ir(110): −13.6, −7.4, and −2.5 kJ mol$^{-1}$, respectively], allowing $O_2$ to adsorb and be further reduced while $H_{ad}$ behaves mainly as a spectator. By contrast, Pt(hkl), especially Pt(111), shows much stronger H-H repulsion (∼ −25 kJ mol$^{-1}$), which more effectively blocks the ensembles required for O–O bond cleavage and thereby favors peroxide formation.[49−51]

In contrast, in alkaline media, $E < 0.30$ V, $H_2O_2$ formation is approximately 21% on Ir(100), followed by 1.6 and 0.8% on Ir(111) and Ir(110), respectively. This result aligns well with the stronger repulsive interaction between $H_{ad}$ and $OH_{ad}$ found on the Ir(100) surface. Above 0.30 V, however, peroxide generation increases substantially, resulting in a different trend in the formation of $H_2O_2$: Ir(111) > Ir(100) > Ir(110). The shift from the 4-electron to 2-electron pathway at higher potentials in alkaline media likely depends on the balance between $OH_{ad}$ and $O_{ad}$ coverage in the peroxide stabilization, as demonstrated by previous DFT calculations.[6] At $E > 0.3$ V, the most important interaction occurs between remaining $OH_{ad}$ and $O_{ad,1}$, conversion of hydroxyl to iridium oxide. From the kinetic model, $\omega_{111} = -30$ kJ mol$^{-1}$, while it is zero for (100) and (110) surfaces, which results in a distinct OH/O co-adsorption dynamic on Ir(111) that likely changes the stabilization of the intermediates for ORR. Note that two surfaces may have similar low-coverage binding energies but very different behavior under reaction conditions. For Ir(*hkl*), this means that peroxide selectivity is not controlled by binding energy alone: lateral interactions determine how rapidly the adsorbed layer destabilizes, how strongly active sites are blocked, and how the stability of key oxygenated intermediates changes with the surface coverage.

For a quantitative analysis, we obtained correlation coefficients between the product of the coverage of oxygenate species (hydroxyl and oxygen) and the peroxide generation in alkaline




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media, as shown in Supplementary Note 4 and Figure S11. A positive correlation of 0.7 for Ir(111) indicates a moderate linear relationship with the degree of coverage of oxygenated species and hydrogen peroxide production. However, for Ir(100), the calculated correlation is –0.3, which results in poor correlation between degree of coverage and $H_2O_2$ generation, further emphasizing the anomalous behavior of this surface. In addition, for Ir(110), the coefficient is –0.9, which indicates a strong negative correlation; that is, the degree of oxygenated species is detrimental for $H_2O_2$ generation, most likely due to its stronger OH/O affinity.

**Oxygen reduction reaction, in acidic media:**

4-Electron pathway (Direct reduction to water):

$$O_2 + 4H^+ + 4\,e^- \rightarrow 2H_2O \tag{10}$$

2-Electron pathway (Reduction to hydrogen peroxide):

$$O_2 + 2H^+ + 2\,e^- \rightarrow H_2O_2 \tag{11}$$

**Oxygen reduction reaction, in alkaline media:**

4-Electron pathway (Direct reduction to water):

$$O_2 + 2H_2O + 4\,e^- \rightarrow 4OH^- \tag{12}$$

2-Electron pathway (Reduction to hydrogen peroxide):

$$O_2 + H_2O + 2\,e^- \rightarrow HO_2^- + OH^- \tag{13}$$

In conclusion, ORR activity is dictated less by oxygen binding energy alone but rather by the surface's ability to stabilize or disrupt a pH-dependent oxygenated interfacial structure that governs both selectivity between the $4e^-$ and $2e^-$ pathways and the reaction kinetics. In alkaline media, surface charge, water orientation, and oxygenated absorbates must therefore be considered as a coupled interfacial ensemble. In this context, $OH_{ad}$ coverage and interfacial water are not merely site-blocking features but active mechanistic contributors, such that $OH_{ad}$ coverage alone cannot reliably predict activity across crystal facets or pH. A complementary mechanistic precedent is provided by Markovic et al.,[52] who highlight that the ORR switches from $4e^-$ to $2e^-$ at higher overpotentials because the hydrogen peroxide reduction reaction (HPRR) becomes kinetically unfavorable, resulting in an apparent net $2e^-$ pathway.

To further infer this mechanistic picture for Ir(*hkl*) surfaces, we examined the HPRR in both acidic and alkaline media. As shown in Figure 3d,e, the HPRR is strongly sensitive to both surface orientation and pH. For clarity, the polarization curves were baseline-corrected by subtracting the corresponding cyclic voltammograms recorded in the absence of $H_2O_2$. The unmodified HPRR polarization curves are provided in the Supporting Information (Figures S12 and S13). In acidic media, the calculated overpotentials at a current density of $-1.5$ mA cm$^{-2}$ are 1.20 V for Ir(100), 1.22 V for Ir(110), and 1.28 V for Ir(111) (Figure 3d). This trend indicates that the Ir(100) and Ir(110) exhibit similar energetics, while Ir(111) is the least active surface. Note that only Ir(100) presents a low-potential inhibition (within the diffusion-limited plateau), which a priori can be related to the stronger repulsive H-OH interaction.

In alkaline media, the overpotential trend shifts distinctly: 1.25 V for Ir(110), 1.35 V for Ir(100), and 1.42 V for Ir(111) (Figure 3f). Importantly, the onset of low-potential inhibition in alkaline conditions tracks the electrode $E_{pztc}$, indicating that the inhibition is governed primarily by surface charge and interfacial water structure rather than simply by $H_{ad}$ adsorption. This interpretation is consistent with recent studies [41,53–55] which identify interfacial water organization and the $E_{pztc}$ as key descriptors of electrocatalytic kinetics. Near the $E_{pztc}$, where the interfacial electric field is minimized, the hydrogen-bond network of water is comparatively more fluid and dynamic. At potentials farther from the $E_{pztc}$, however, the stronger interfacial field induces greater orientational ordering, leading to a more rigid water network that can impede the interfacial transport steps coupled to electrocatalysis. Mechanistically, the comparison between ORR and HPRR in alkaline media suggests that at higher potentials ($E > 0.4$ V) HPRR becomes inhibited, promoting $H_2O_2$ accumulation during ORR and thus appearing as a pathway shift. However, the current inhibition mismatch between HPRR and ORR, for instance, $E > 0.4$ V in acidic media, implies that ORR does not proceed exclusively through $H_2O_2$ as an obligatory intermediate; otherwise, ORR inhibition would be expected to mirror HPRR inhibition more closely.

**Hydrogen peroxide reduction reaction, in acidic media**

$$H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O \tag{14}$$

**Hydrogen peroxide reduction reaction, in alkaline media:**

$$HO_2^- + H_2O + 2e^- \rightarrow 3OH^- \tag{15}$$

## General Trends and Trade-Offs across Reactions, pH, and the Surface Structure

Across HER, HOR, ORR, and HPRR on Ir(111), Ir(100), and Ir(110), the main outcome is that activity is both reaction-specific and facet-dependent, and pH amplifies these differences. Consistent with the polarization metrics, the turnover frequency (TOF) trends shown in Table 1 (for calculation details, see Supplementary Note 5 and Table S4 ) reinforce that no single facet maximizes turnover across all reactions and electrolytes. This reflects the competing requirements, such as lateral interaction among adjacent co-adsorbates, that will directly influence the coverage degree (active site availability) and reactants/ intermediates/products binding energy (spectators vs site-blocking species and change in mechanism pathway), and stabilization via interfacial water structure (altering kinetics descriptors), as shown in Figure 4.

For hydrogen reactions (HER/HOR), kinetic parameters, such as overpotential and current density at fixed potential, demonstrated Ir(111) as the most active surface in both acidic and alkaline media, with Ir(110) intermediate and Ir(100) least active. However, TOF analysis (Table 1) shows a change in the trend (acidic media) as follows: Ir(110) > Ir(111) > Ir(100), a result of the higher atomic density of the (110) surface. This effect becomes even more pronounced when comparing the unreconstructed ($1 \times 1$) surfaces with the reconstructed phases. In contrast, in alkaline media, the differences in HER/HOR activity among the Ir(*hkl*) facets are sufficiently large that the overall activity trend appears insensitive to the specific atomic arrangement, with Ir(111) remaining the most active surface. The dominant trade-off is pH: moving from acid to base causes

**Table 1. Turnover Frequency (TOF) and corresponding standard deviation from repeat experiments ($n$ > 3) on different iridium single-crystal surfaces for HER, HOR, ORR, and HPRR in 0.1 M $HClO_4$ and 0.1 M KOH, Including Reconstructed Surfaces for Ir(100) and Ir(110) in Dashed Bars[a]**

| Materials | | TOF ($s^{-1}$) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Acidic (0.1 M $HClO_4$) | | | | Alkaline (0.1 M KOH) | | | |
| Facet | Surface Sites | HER | HOR | ORR | HPRR | HER | HOR | ORR | HPRR |
| Ir(111) | (1 × 1) | 13.3 ± 3.5 | 4.8 ± 0.4 | 0.7 ± 0.2 | 2.9 ± 0.3 | 3.0 ± 0.8 | 2.3 ± 0.3 | 1.4 ± 0.5 | 0.7 ± 0.1 |
| Ir(100) | (1 × 1) | 13.2 ± 2.0 | 5.2 ± 0.4 | 3.7 ± 0.1 | 3.98 ± 0.01 | 0.2 ± 0.1 | 0.9 ± 0.4 | 3.3 ± 0.3 | 2.7 ± 0.1 |
| | (5 × 1) (hex) | 11.0 ± 1.7 | 4.3 ± 0.4 | 3.1 ± 0.1 | 3.3 ± 0.01 | 0.2 ± 0.1 | 0.7 ± 0.4 | 2.7 ± 0.3 | 2.3 ± 0.1 |
| Ir(110) | 1 × 1 | 13.7 ± 4.2 | 5.00 ± 1.5 | 6.2 ± 1.6 | 4.9 ± 0.8 | 0.7 ± 0.2 | 0.9 ± 0.2 | 5.1 ± 0.8 | 1.6 ± 0.4 |
| | (1 × 2) (MR) | 27.4 ± 8.4 | 10.0 ± 3.0 | 12.3 ± 3.2 | 9.8 ± 1.7 | 1.3 ± 0.3 | 1.9 ± 0.3 | 10.3 ± 1.5 | 3.2 ± 0.9 |

[a]TOF values are obtained from $j_{HER}$ at −20 mV, $j_{HOR}$ at 20 mV, $j_{ORR}$ and $j_{HPRR}$ at 500 mV.

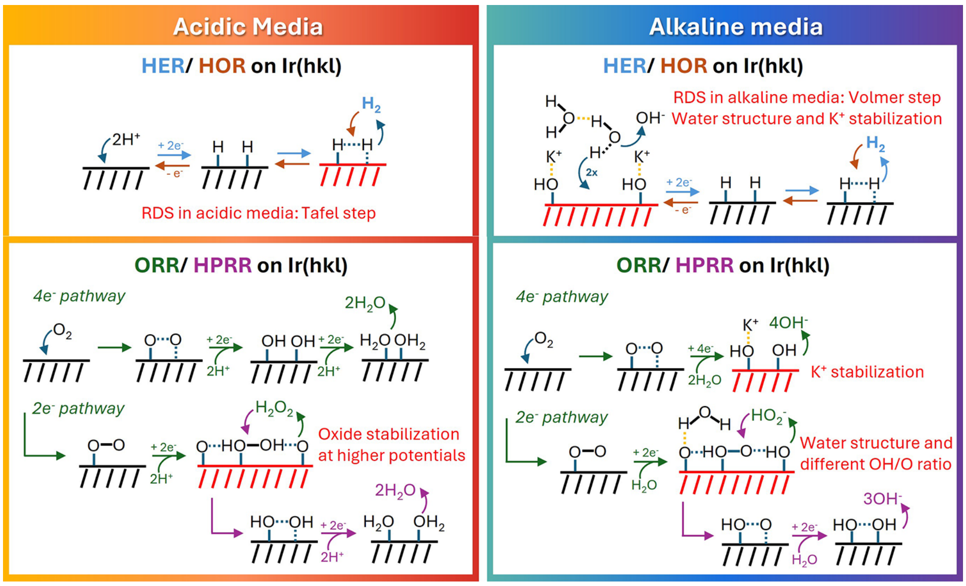


**Figure 4.** Chemical diagram of the electrocatalytic reactions and parameters that affect the activity and selectivity.

a large TOF suppression on all facets, consistent with an added kinetic burden from water dissociation and double layer/water reorganization. Thus, surfaces that may bind $H_{ad}$ strongly (beneficial for forming H*) can still exhibit lower TOF if that binding promotes spectator $H_{ad}/OH_{ad}$ coverage and reduces the population of reactive sites.

For the oxygen reduction reaction, TOF follows the same qualitative structure sensitivity as the kinetic overpotentials: Ir(110) shows the highest ORR turnover regardless of pH, while Ir(111) remains the least active, independently if a surface reconstruction takes place on (100) or (110) surfaces (Table 1). In contrast to the HER/HOR, ORR displays a comparatively weak pH dependence in activity/TOF, indicating that facet-dependent oxygen activation and oxygenate stabilization dominate over proton-availability effects.

Finally, hydrogen peroxide reactions introduce a cross-reaction constraint. While ORR in acidic media yields very little peroxide formation, alkaline conditions strongly increase peroxide production. Therefore, the effectiveness of HPRR as a mitigation pathway becomes critical. The HPRR TOF trends show that Ir(111) is consistently the slowest surface (Table 1). Moreover, in acidic media, the trends are independent of iridium phases: Ir(110) > Ir(100) > Ir(111). In contrast, in alkaline media, the surface reconstruction plays a role: considering the unreconstructed surfaces, (100) > (110) > (111), while reconstructed surfaces alter the trend: (110) ≈ (100) > (111). This pH-dependent switch highlights a key trade-off: a surface can be ORR-active yet still allows peroxide accumulation if HPRR turnover is inhibited in the relevant potential window (linked to $E_{pztc}$/interfacial water effects), meaning that selectivity and stability cannot be inferred from ORR activity alone. Overall, Table 1 provides a unified view: (111) is the best "hydrogen facet", (110) is the best "oxygen-reduction facet", and peroxide consumption (HPRR) is both facet- and pH-dependent, making catalyst optimization intrinsically multi-objective and environment-specific. Figure 4 summarizes the most noticeable mechanistic aspects for HER, HOR, ORR, and HPRR in both acidic and alkaline media, highlighting the important parameters discussed in this work.

## ■ CONCLUSIONS

This work establishes benchmark structure-reactivity relationships for HER, HOR, ORR, and HPRR on Ir(111), Ir(100), and Ir(110) in acidic and alkaline media. All reactions show strong facet and pH dependence, demonstrating that iridium electrocatalysis cannot be described by surface coordination alone, but instead reflects the coupled effects of adsorption




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energetics, adsorbate coverage, lateral interactions, and interfacial water structure.

For hydrogen electrocatalysis, Ir(111) is the most active facet in both media, whereas for ORR the activity trend is reversed and Ir(110) is the most active. ORR selectivity is highly pH-dependent: $H_2O_2$ formation is negligible in acidic media but increases strongly in alkaline media, where it correlates with oxygenated surface species and inhibited HPRR. These results indicate that peroxide production on Ir is governed not only by oxygen binding, but also by pH-dependent interfacial structure and water-mediated processes.

Turnover-frequency analysis confirms the strong facet-dependent activity trends for hydrogen reactions and the relative pH invariance of ORR rates, while also showing that peroxide consumption (HPRR) is as facet-sensitive as peroxide formation. Collectively, these results highlight that evaluating and thus designing electrocatalysts requires simultaneously accounting for kinetics, selectivity (4-electron vs 2-electron ORR), peroxide generation and removal, coverage/poisoning effects, and pH-dependent interfacial dynamics; otherwise, conclusions drawn from a single reaction, potential window, or metric (overpotential, current density, or TOF alone) can be misleading for real operating environments.

## ■ EXPERIMENTAL SECTION

### Surface Preparation

The working electrodes used in this study were iridium low-index single-crystalline surfaces, specifically Ir(111), Ir(100), and Ir(110). Each electrode had an estimated geometric area of 0.283 $cm^2$. To ensure surface cleanliness and structural integrity, the electrodes were prepared as described previously.[29,32,36,37,56] Shortly after, it was annealed at 1600 °C in a 3% $H_2$/Ar atmosphere for 10 min. After annealing, the electrodes were allowed to cool for 15 min, during which 1–2 drops of ultrapure deionized water (Milli Q, 18.2 MΩ cm) were added on top of the surface to prevent air exposure and contamination. The electrodes were then inverted over a clean and water boiled prolene film sitting over a flat and polished PTFE plate, and assembled into the rotating disk electrode (RDE) or rotating ring disk electrode (RRDE) setup with use of a PTFE sleeve (U-cup), transferred to the electrochemical cell, and fully submerged in the electrolyte while polarized at 0.05 V vs the reversible hydrogen electrode (RHE).

### AFM Measurements

Atomic force microscopy (AFM) measurements were performed using a Bruker Dimension FastScan instrument operated in tapping mode under ambient conditions. Surface topography images were acquired on freshly annealed surfaces and on surfaces following immersion in 0.1 M $HClO_4$ at a potential of 0.05 V versus the reversible hydrogen electrode (RHE). Scan areas ranged from 500 nm × 500 nm to 5 μm × 5 μm. The scan resolution was set to 256 lines for 500 nm scans and 1024 lines for 2 μm scans. A scan rate of 1 Hz and a scanning angle of 0° were used throughout. The *Z* limit was maintained at 2 μm to ensure adequate height sensitivity. Raw AFM data were processed and visualized using Gwyddion software. A third-degree polynomial function was applied to perform line alignment and background correction, thereby removing residual tilt and curvature artifacts from the images. Line profiles were extracted across the images to evaluate step edge morphology and terrace structure. Mean roughness and root-mean-square roughness were calculated from the images to quantify topographic changes between pristine and electrolyte-immersed surfaces. All measurements were conducted under consistent ambient conditions to ensure reproducibility of the acquired data.

### Electrochemical Measurements

All electrochemical experiments were conducted in a three-electrode system housed within a well-enclosed fluorinated ethylene propylene (FEP) cell. A graphite rod (99.995% purity) or an Ir wire was used as the counter electrode. The reference electrode was an Ag/AgCl electrode connected to the main compartment of the cell via a glass bridge (calibrated using a RHE electrode in the corresponding electrolyte). In this investigation, all potentials were referenced to RHE. Suitable ohmic drop corrections were applied to all measurements to ensure accurate data interpretation. Electrolyte solutions were prepared by diluting perchloric acid ($HClO_4$, Omnitrace Ultra, EMD) in ultrapure Milli-Q water or by dissolving Suprapur grade potassium hydroxide hydrate (99.995%, Merck) in ultrapure Milli-Q water. The measured pH of 0.1 M $HClO_4$ solution was 1.02, and the 0.1 M KOH solution was 13.01. Additionally, in this work, all density currents were calculated assuming the geometric area of the single-crystal surfaces. This assumption is based on the presence of large terraces, resulting in a low atomic roughness, as shown in AFM analysis (Figure S14, Supplementary Note 5).

Electrochemical measurements were performed using an Autolab PGSTAT302N potentiostat equipped with a Scan 250 module. Cyclic voltammograms (CVs) were recorded at a sweep rate of 50 mV $s^{-1}$ unless otherwise specified. Ohmic drop correction was applied in all electrochemical measurements. A correction of 23 Ω was used for experiments conducted in 0.1 M $HClO_4$, and a correction of 35 Ω was used for experiments in 0.1 M KOH. The uncompensated resistance was estimated using the current interrupt method, and 96–97% of the ohmic drop was corrected in each case. Prior to each experiment, the electrochemical cell was purged with high-purity argon gas (99.9999%, Airgas) to remove dissolved oxygen from the electrolyte. During data collection, the argon flow was maintained at a reduced rate to minimize disturbances. Depending on the experimental requirements, measurements were carried out in either $O_2$-saturated (Airgas, 99.9999% Research Plus grade), $H_2$-saturated (Airgas, Research Plus grade, 99.9999%), or Ar-purged electrolytes.

The RDE and RRDE setups were employed to evaluate the electrocatalytic performance of Ir surfaces. For RRDE experiments, the ring electrode was polarized at 1.2 V to detect hydrogen peroxide formed during oxygen reduction reaction (ORR) measurements. The RRDE setup was composed of a gold ring (7.5 mm inner diameter, 8.5 mm outer diameter) embedded within the collet that holds the disk electrode and its PTFE sleeve (U-cup). Before every experiment, the ring was polished using a 0.05 μm $Al_2O_3$ slurry on a microfiber polishing pad and cleaned by sonicating for 5 min in isopropanol, followed by sonication in ultrapure water twice. All data were collected under controlled rotation speeds and scan rates to ensure reproducibility and consistency across experiments.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.6c02938.

> CO-displacement data, calculation of potential zero total charge, supplementary figures and tables, microkinetic adsorption simulations, correlation between surface coverage and peroxide formation, AFM data, and TOF calculations (DOCX)

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### Author Contributions

P.P.L. and K.A. conceptualized the study. K.A. performed the electrochemical studies. K.A. and K.N.S. performed the kinetic analysis. K.A., K.N.S., and P.P.L. drafted the manuscript, and all authors approved its final version.

### Notes

The authors declare no competing financial interest.

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## ■ REFERENCES


(1) Lopes, P. P.; Strmcnik, D.; Tripkovic, D.; Connell, J. G.; Stamenkovic, V.; Markovic, N. M. Relationships between Atomic Level Surface Structure and Stability/Activity of Platinum Surface Atoms in Aqueous Environments. *ACS Catal.* **2016**, *6* (4), 2536–2544.

(2) Lopes, P. P.; Li, D.; Lv, H.; Wang, C.; Tripkovic, D.; Zhu, Y.; Schimmenti, R.; Daimon, H.; Kang, Y.; Snyder, J.; Becknell, N.; More, K. L.; Strmcnik, D.; Markovic, N. M.; Mavrikakis, M.; Stamenkovic, V. R. Eliminating Dissolution of Platinum-Based Electrocatalysts at the Atomic Scale. *Nat. Mater.* **2020**, *19* (11), 1207–1214.

(3) Lopes, P. P.; Tripkovic, D.; Martins, P. F. B. D.; Strmcnik, D.; Ticianelli, E. A.; Stamenkovic, V. R.; Markovic, N. M. Dynamics of Electrochemical Pt Dissolution at Atomic and Molecular Levels. *J. Electroanal. Chem.* **2018**, *819*, 123–129.

(4) Dong, J. C.; Zhang, X. G.; Briega-Martos, V.; Jin, X.; Yang, J.; Chen, S.; Yang, Z. L.; Wu, D. Y.; Feliu, J. M.; Williams, C. T.; Tian, Z. Q.; Li, J. F. In Situ Raman Spectroscopic Evidence for Oxygen Reduction Reaction Intermediates at Platinum Single-Crystal Surfaces. *Nat. Energy* **2019**, *4* (1), 60–67.

(5) Maciá, M. D.; Campiña, J. M.; Herrero, E.; Feliu, J. M. On the Kinetics of Oxygen Reduction on Platinum Stepped Surfaces in Acidic Media. *J. Electroanal. Chem.* **2004**, *564* (1–2), 141–150.

(6) Nascimento da Silva, K.; Anjanarambath, K.; Williams, C. K.; He, H.; Zapol, P.; Lopes, P. P. Ir(Hkl) Surface Electrochemistry in a Nonadsorbing Acidic Medium. *J. Am. Chem. Soc.* **2025**, *147* (34), 30708–30722.

(7) Stamenkovic, V. R.; Strmcnik, D.; Lopes, P. P.; Markovic, N. M. Energy and Fuels from Electrochemical Interfaces. *Nat. Mater.* **2016**, *16* (1), 57–69.

(8) Montoya, J. H.; Seitz, L. C.; Chakthranont, P.; Vojvodic, A.; Jaramillo, T. F.; Nørskov, J. K. Materials for Solar Fuels and Chemicals. *Nat. Mater.* **2016**, *16* (1), 70–81.

(9) Shao, M.; Chang, Q.; Dodelet, J. P.; Chenitz, R. Recent Advances in Electrocatalysts for Oxygen Reduction Reaction. *Chem. Rev.* **2016**, *116* (6), 3594–3657.

(10) Dey, S.; Mondal, B.; Chatterjee, S.; Rana, A.; Amanullah, S.; Dey, A. Molecular Electrocatalysts for the Oxygen Reduction Reaction. *Nat. Rev. Chem.* **2017**, *1* (12), 0098.

(11) Li, Y.; Li, Q.; Wang, H.; Zhang, L.; Wilkinson, D. P.; Zhang, J. Recent Progresses in Oxygen Reduction Reaction Electrocatalysts for Electrochemical Energy Applications. *Electrochem. Energy Rev.* **2019**, *2* (4), 518–538.

(12) Sitta, E.; Gómez-Marín, A. M.; Aldaz, A.; Feliu, J. M. Electrocatalysis of $H_2O_2$ Reduction/Oxidation at Model Platinum Surfaces. *Electrochem. Commun.* **2013**, *33*, 39–42.

(13) Marković, N. M.; Schmidt, T. J.; Stamenković, V.; Ross, P. N. Oxygen Reduction Reaction on Pt and Pt Bimetallic Surfaces: A Selective Review. *Fuel Cells* **2001**, *1* (2), 105–116.

(14) Paulus, U. A.; Schmidt, T. J.; Gasteiger, H. A.; Behm, R. J. Oxygen Reduction on a High-Surface Area Pt/Vulcan Carbon Catalyst: A Thin-Film Rotating Ring-Disk Electrode Study. *J. Electroanal. Chem.* **2001**, *495*, 134–145.

(15) Kinumoto, T.; Inaba, M.; Nakayama, Y.; Ogata, K.; Umebayashi, R.; Tasaka, A.; Iriyama, Y.; Abe, T.; Ogumi, Z. Durability of Perfluorinated Ionomer Membrane against Hydrogen Peroxide. *J. Power Sources* **2006**, *158*, 1222–1228.

(16) Inaba, M.; Kinumoto, T.; Kiriake, M.; Umebayashi, R.; Tasaka, A.; Ogumi, Z. Gas Crossover and Membrane Degradation in Polymer Electrolyte Fuel Cells. *Electrochim. Acta* **2006**, *51* (26), 5746–5753.

(17) Sitta, E.; Da Silva, K. N.; Feliu, J. M. Hydrogen Peroxide Oxidation/Reduction Reaction on Platinum Surfaces. In *Encyclopedia of Interfacial Chemistry: Surface Science and Electrochemistry*; Elsevier, **2018**; pp 682–689.

(18) Koper, M. T. M. *Fuel Cell Catalysis. A Surface Science Approach*; Wiley, **2009**.

(19) Viswanathan, V.; Hansen, H. A.; Rossmeisl, J.; Nørskov, J. K. Unifying the $2e^-$ and $4e^-$ Reduction of Oxygen on Metal Surfaces. *J. Phys. Chem. Lett.* **2012**, *3* (20), 2948–2951.

(20) Messias, I.; Kupferberg, J.; Bielinski, A. R.; Nagao, R.; Martinson, A. B. F.; Papa Lopes, P. Quantification of Reactive Oxygen Species Produced from Electrocatalytic Materials. *ACS Catal.* **2025**, *15* (4), 2750–2759.

(21) Mo, T.; Zalitis, C. M.; Jackson, C.; Petrucco, E.; Sharman, J.; Kucernak, A. R. J. From Formation to Failure: The Role of Hydrogen Peroxide in Proton Exchange Membrane Technologies. *ACS Catal.* **2026**, *16*, 4266–4289.

(22) Johnson, E.; Mukundan, R.; Weber, A. Z. Membrane Degradation in PEM Fuel Cells: Part I. Modeling Gas Crossover and the Pt Band. *J. Electrochem. Soc.* **2026**, *173*, 024504.

(23) Borup, R.; Meyers, J.; Pivovar, B.; Kim, Y. S.; Mukundan, R.; Garland, N.; Myers, D.; Wilson, M.; Garzon, F.; Wood, D.; Zelenay, P.; More, K.; Stroh, K.; Zawodzinski, T.; Boncella, J.; McGrath, J. E.; Inaba, M.; Miyatake, K.; Hori, M.; Ota, K.; Ogumi, Z.; Miyata, S.; Nishikata, A.; Siroma, Z.; Uchimoto, Y.; Yasuda, K.; Kimijima, K. I.; Iwashita, N. Scientific Aspects of Polymer Electrolyte Fuel Cell Durability and Degradation. *Chem. Rev.* **2007**, *107* (10), 3904–3951.

(24) Van Hove, M. A.; Weinberg, W. H.; Chan, C.-M. *Low-Energy Electron Diffraction*; Springer: Berlin, Heidelberg, **1986**; Vol. *6*.

(25) Wei, J.; Zheng, Y. L.; Li, Z. Y.; Xu, M. Le; Chen, Y. X.; Ye, S. Electrochemistry of Oxygen at Ir Single Crystalline Electrodes in Acid. *Electrochim. Acta* **2017**, *246*, 329–337.

(26) Gómez, R.; Weaver, M. J. Reduction of Nitrous Oxide on Iridium Single-Crystal Electrodes. *Langmuir* **2002**, *18* (11), 4426–4432.

(27) Wan, L. J.; Hara, M.; Inukai, J.; Itaya, K. In Situ Scanning Tunneling Microscopy of Well-Defined Ir(111) Surface: High-Resolution Imaging of Adsorbed Sulfate. *J. Phys. Chem. B* **1999**, *103* (33), 6978–6983.

(28) Motoo, S.; Furuya, N. Effect of Anions on Hydrogen and Oxygen Adsorption on Iridium Single Crystal Surfaces. *J. Electroanal. Chem* **1984**, *181*, 301–305.

(29) Pajkossy, T.; Kibler, L. A.; Kolb, D. M. Voltammetry and Impedance Measurements of Ir(1 1 1) Electrodes in Aqueous Solutions. *J. Electroanal. Chem.* **2005**, *582* (1–2), 69–75.

(30) Hayashi, K.; Tomimori, T.; Chida, Y.; Todoroki, N.; Wadayama, T. Hydrogen Peroxide Generation and Hydrogen Oxidation Reaction on Pt/Co/Pt(111) and Pt/Co/Pt(100) Single-Crystal Model Catalyst Surface. *ACS Catal.* **2024**, *14* (18), 14069–14077.

(31) Todoroki, N.; Watanabe, H.; Kondo, T.; Kaneko, S.; Wadayama, T. Highly Enhanced Oxygen Reduction Reaction Activity and Electrochemical Stability of Pt/Ir(111) Bimetallic Surfaces. *Electrochim. Acta* **2016**, *222*, 1616–1621.

(32) Lopes, P. P.; Strmcnik, D.; Jirkovsky, J. S.; Connell, J. G.; Stamenkovic, V.; Markovic, N. Double Layer Effects in Electrocatalysis: The Oxygen Reduction Reaction and Ethanol Oxidation Reaction on Au(1 1 1), Pt(1 1 1) and Ir(1 1 1) in Alkaline Media Containing Na and Li Cations. *Catal. Today* **2016**, *262*, 41–47.

(33) Strmcnik, D.; Uchimura, M.; Wang, C.; Subbaraman, R.; Danilovic, N.; Van Der Vliet, D.; Paulikas, A. P.; Stamenkovic, V. R.; Markovic, N. M. Improving the Hydrogen Oxidation Reaction Rate by Promotion of Hydroxyl Adsorption. *Nat. Chem.* **2013**, *5* (4), 300–306.

(34) Strmcnik, D.; Kodama, K.; Van Der Vliet, D.; Greeley, J.; Stamenkovic, V. R.; Marković, N. M. The Role of Non-Covalent Interactions in Electrocatalytic Fuel-Cell Reactions on Platinum. *Nat. Chem.* **2009**, *1* (6), 466–472.

(35) Damjanovic, A.; Yeh, L.-S. R.; Wolf, J. F. Formation of Thin Oxide Films at Platinum Anodes in Alkaline Solutions: III. PH Dependence. *J. Electrochem. Soc.* **1982**, *129*, 55–61.

(36) Pajkossy, T.; Kibler, L. A.; Kolb, D. M. Voltammetry and Impedance Measurements of Ir(1 0 0) Electrodes in Aqueous Solutions. *J. Electroanal. Chem.* **2007**, *600* (1), 113–118.

(37) Ganassin, A.; Sebastian, P.; Climent, V.; Schuhmann, W.; Bandarenka, A. S.; Feliu, J. On the PH Dependence of the Potential of Maximum Entropy of Ir(111) Electrodes. *Sci. Rep.* **2017**, *7* (1), 1246.

(38) Strmcnik, D.; Lopes, P. P.; Genorio, B.; Stamenkovic, V. R.; Markovic, N. M. Design Principles for Hydrogen Evolution Reaction Catalyst Materials. *Nano Energy* **2016**, *29*, 29–36.

(39) Khadke, P.; Tichter, T.; Boettcher, T.; Muench, F.; Ensinger, W.; Roth, C. A Simple and Effective Method for the Accurate Extraction of Kinetic Parameters Using Differential Tafel Plots. *Sci. Rep.* **2021**, *11* (1), 8974.

(40) van der Heijden, O.; Park, S.; Vos, R. E.; Eggebeen, J. J. J.; Koper, M. T. M. Tafel Slope Plot as a Tool to Analyze Electrocatalytic Reactions. *ACS Energy Lett.* **2024**, *9* (4), 1871–1879.

(41) Ledezma-Yanez, I.; Wallace, W. D. Z.; Sebastián-Pascual, P.; Climent, V.; Feliu, J. M.; Koper, M. T. M. Interfacial Water Reorganization as a PH-Dependent Descriptor of the Hydrogen Evolution Rate on Platinum Electrodes. *Nat. Energy* **2017**, *2* (4), 17031.

(42) Chunli, L.; Ling, Z.; Xiaodong, W.; Yong, Y.; Yong-Wang, L.; Haijun, J. Hydrogen Adsorption on Ir(111), Ir(100) and Ir(110)—Surface and Coverage Dependence. *Surf. Sci.* **2020**, *692*, 121514.

(43) Lee, S. H.; Rusakova, I.; Hoffman, D. M.; Jacobson, A. J.; Lee, T. R. Monodisperse $SnO_2$-Coated Gold Nanoparticles Are Markedly More Stable than Analogous $SiO_2$-Coated Gold Nanoparticles. *ACS Appl. Mater. Interfaces* **2013**, *5* (7), 2479–2484.

(44) Durst, J.; Siebel, A.; Simon, C.; Hasché, F.; Herranz, J.; Gasteiger, H. A. New Insights into the Electrochemical Hydrogen Oxidation and Evolution Reaction Mechanism. *Energy Environ. Sci.* **2014**, *7* (7), 2255–2260.

(45) Bard, A. J.; Faulkner, L. R. *Electrochemical Methods : Fundamentals and Applications*; John Wiley & Sons, Inc., **2001**.

(46) Lima, F. H. B.; Zhang, J.; Shao, M. H.; Sasaki, K.; Vukmirovic, M. B.; Ticianelli, E. A.; Adzic, R. R. Catalytic Activity - d-Band Center Correlation for the $O_2$ Reduction Reaction on Platinum in Alkaline Solutions. *J. Phys. Chem. C* **2007**, *111* (1), 404–410.

(47) Hayashi, K.; Tomimori, T.; Chida, Y.; Todoroki, N.; Wadayama, T. Hydrogen Peroxide Generation and Hydrogen Oxidation Reaction Properties of Ir(111)-, (100)-, and (110)-Low-Index Single-Crystal Surfaces. *J. Phys. Chem. C* **2021**, *125* (39), 21481–21487.

(48) Stamenkovic, V. R.; Fowler, B.; Mun, B. S.; Wang, G.; Ross, P. N.; Lucas, C. A.; Markovic, N. M. Improved Oxygen Reduction Activity on $Pt_3Ni(111)$ via Increased Surface Site Availability. *Science* **2007**, *315* (5811), 493–497.

(49) Markovic, N.; Gasteiger, H.; Ross, P. N. Kinetics of Oxygen Reduction on Pt(Hkl) Electrodes: Implications for the Crystallite Size Effect with Supported Pt Electrocatalysts. *J. Electrochem. Soc.* **1997**, *144*, 1591.

(50) Gómez, R.; Orts, J. M.; Álvarez-Ruiz, B.; Feliu, J. M. Effect of Temperature on Hydrogen Adsorption on Pt(111), Pt(110), and Pt(100) Electrodes in 0.1 M $HClO_4$. *J. Phys. Chem. B* **2004**, *108* (1), 228–238.

(51) Liu, J.; Hagopian, A.; McCrum, I. T.; Doblhoff-Dier, K.; Koper, M. T. M. Unraveling the Origin of the Repulsive Interaction between Hydrogen Adsorbates on Platinum Single-Crystal Electrodes. *J. Phys. Chem. C* **2024**, *128* (36), 15019–15028.

(52) Markovic, N. M.; Tidswell, I. M.; Ross, P. N. Oxygen and Hydrogen Peroxide Reduction on the gold(100) Surface in Alkaline Electrolyte: The Roles of Surface Structure and Hydroxide Adsorption. *Langmuir* **1994**, *10* (1), 1–4.

(53) Ramaswamy, N.; Ghoshal, S.; Bates, M. K.; Jia, Q.; Li, J.; Mukerjee, S. Hydrogen Oxidation Reaction in Alkaline Media: Relationship between Electrocatalysis and Electrochemical Double-Layer Structure. *Nano Energy* **2017**, *41*, 765–771.

(54) Shen, L.-f.; Lu, B.-a.; Li, Y.-y.; Liu, J.; Huang-fu, Z.-c.; Peng, H.; Ye, J.-y.; Qu, X.-m.; Zhang, J.-m.; Li, G.; Cai, W.-b.; Jiang, Y.-x.; Sun, S.-g. Interfacial Structure of Water as a New Descriptor of the Hydrogen Evolution Reaction. *Angew. Chem., Int. Ed.* **2020**, *59* (50), 22397–22402.

(55) Li, P.; Jiang, Y.; Hu, Y.; Men, Y.; Liu, Y.; Cai, W.; Chen, S. Hydrogen Bond Network Connectivity in the Electric Double Layer Dominates the Kinetic PH Effect in Hydrogen Electrocatalysis on Pt. *Nat. Catal.* **2022**, *5* (10), 900–911.

(56) Özer, E.; Spöri, C.; Reier, T.; Strasser, P. Iridium(1 1 1), Iridium(1 1 0), and Ruthenium(0 0 0 1) Single Crystals as Model Catalysts for the Oxygen Evolution Reaction: Insights into the Electrochemical Oxide Formation and Electrocatalytic Activity. *ChemCatChem* **2017**, *9* (4), 597–603.





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